CompTIA Security+ SY0-701 Certification Exam Objectives

Bloom's Taxonomy is an industry-standard classification system used to help identify the level of ability that learners need to demonstrate proficiency. It is often used to classify educational learning objectives into different levels of complexity. The Bloom's Taxonomy column in the following table reflects the level of coverage for the respective SY0-701 objective domains. In all instances, the level of coverage in *CompTIA Security+ Guide to Network Security Fundamentals, 8th Edition* meets or exceeds that indicated by CompTIA for that objective. See the Introduction of this book for more information.

Security+ Exam Objective Domain/Objectives	Module	Section	Bloom's Taxonomy
1.0 General Security Concepts			
1.1 Compare and contrast various types of security controls.			
• Categories ○ Technical ○ Managerial ○ Operational ○ Physical	1	How Attacks Occur	Remembering
• Control types ○ Preventive ○ Deterrent ○ Detective ○ Corrective ○ Compensating ○ Directive		What Is Information Security?	Understanding
1.2 Summarize fundamental security concepts.			
• Confidentiality, Integrity, and Availability (CIA) • Non-repudiation • Authentication, Authorization, and Accounting (AAA) ○ Authenticating people ○ Authenticating systems ○ Authorization models	1	What Is Information Security?	Applying
• Gap analysis	9	Security Appliances	Understanding
• Zero trust ○ Control plane ▪ Adaptive identity ▪ Threat scope reduction ▪ Policy-driven access control ▪ Secured zones ○ Data plane ▪ Subject/system ▪ Policy engine ▪ Policy automation ▪ Policy enforcement point	9	Secure Infrastructure Design	Applying
• Physical security ○ Bollards ○ Access control vestibule ○ Fencing ○ Video surveillance ○ Security guard ○ Access badge ○ Lighting ○ Sensors ▪ Infrared ▪ Pressure ▪ Microwave ▪ Ultrasonic	2	Physical Security Controls	Applying
• Deception and disruption technology ○ Honeypot ○ Honeynet ○ Honeyfile ○ Honeytoken	9	Infrastructure Security Hardware	Understanding
1.3 Explain the importance of change management processes and the impact to security.			
• Business processes impacting security operation ○ Approval process ○ Ownership ○ Stakeholders ○ Impact analysis ○ Test results	15	Asset Protection	Analyzing

◦ Backout plan ◦ Maintenance window ◦ Standard operating procedure • Technical implications ◦ Allow lists/deny lists ◦ Restricted activities ◦ Downtime ◦ Service restart ◦ Application restart ◦ Legacy applications ◦ Dependencies • Documentation ◦ Updating diagrams ◦ Updating policies/procedures • Version control			
1.4 Explain the importance of using appropriate cryptographic solutions.			
• Public key infrastructure (PKI) ◦ Public key ◦ Private key ◦ Key escrow	4	Public Key Infrastructure (PKI)	Applying
• Encryption ◦ Level ▪ Full-disk ▪ Partition ▪ File ▪ Volume ▪ Database ▪ Record ◦ Transport/communication ◦ Asymmetric ◦ Symmetric ◦ Key exchange ◦ Algorithms ◦ Key length	3	Defining Cryptography	Evaluating
• Tools ◦ Trusted Platform Module (TPM) ◦ Hardware security module (HSM) ◦ Key management system ◦ Secure enclave	3	Using Cryptography	Applying
• Obfuscation ◦ Steganography ◦ Tokenization ◦ Data masking	3	Defining Cryptography	Analyzing
• Hashing	3	Cryptographic Algorithms	Understanding
• Salting	7	Authentication Best Practices	Evaluating
• Digital signatures • Key stretching	3	Using Cryptography	Creating
• Blockchain • Open public ledger			
• Certificates ◦ Certificate authorities ◦ Certificate revocation lists (CRLs) ◦ Online Certificate Status Protocol (OCSP) ◦ Self-signed ◦ Third-party ◦ Root of trust ◦ Certificate signing request (CSR) generation ◦ Wildcard	4	Digital Certificates	Applying
2.0 Threats, Vulnerabilities, and Mitigations			
2.1 Compare and contrast common threat actors and motivations.			

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○ Nation-state ○ Unskilled attacker ○ Hacktivist ○ Insider threat ○ Organized crime ○ Shadow IT • Attributes of actors ○ Internal/external ○ Resources/funding ○ Level of sophistication/capability • Motivations ○ Data exfiltration ○ Espionage ○ Service disruption ○ Blackmail ○ Financial gain ○ Philosophical/political beliefs ○ Ethical ○ Revenge ○ Disruption/chaos ○ War		Motivations	
2.2 Explain common threat vectors and attack surfaces. • Message-based ○ Email ○ Short Message Service (SMS) ○ Instant messaging (IM) • Image-based • File-based • Voice call • Removable device • Vulnerable software ○ Client-based vs. agentless • Unsupported systems and applications	1	How Attacks Occur	Applying
• Unsecure networks ○ Wireless ○ Wired	10	Wireless Attacks	Analyzing
○ Bluetooth • Open service ports • Default credentials • Supply chain ○ Managed service providers (MSPs) ○ Vendors ○ Suppliers	1	How Attacks Occur	Understanding
• Human vectors/social engineering ○ Phishing ○ Vishing ○ Smishing ○ Misinformation/disinformation ○ Impersonation ○ Business email compromise ○ Pretexting ○ Watering hole ○ Brand impersonation ○ Typo squatting	2	Social Engineering Attacks	Creating
2.3 Explain various types of vulnerabilities. • Application ○ Memory injection ○ Buffer overflow ○ Race conditions ▪ Time-of-check (TOC) ▪ Target of evaluation (TOE) ▪ Time-of-use (TOU)	5	Application Vulnerabilities and Attacks	Understanding

○ Malicious update			
• Operating system (OS)–based	1	How Attacks Occur	Remembering
• Web-based ○ Structured Query Language injection (SQLi) ○ Cross-site scripting (XSS)	5	Application Vulnerabilities and Attacks	Applying
• Hardware ○ Firmware ○ End-of-life ○ Legacy	1	How Attacks Occur	Applying
• Virtualization ○ Virtual machine (VM) escape ○ Resource reuse	11	Virtualization Security	Remembering
• Cloud-specific	11	Cloud Computing Security	Understanding
• Supply chain ○ Service provider ○ Hardware provider ○ Software provider	1	How Attacks Occur	
• Cryptographic	3	Fundamentals of Cryptography	Remembering
• Misconfiguration			
• Mobile device ○ Side loading ○ Jailbreaking	6 1	Mobile Device Risks How Attacks Occur	Understanding Applying
• Zero-day			
2.4 Given a scenario, analyze indicators of malicious activity.			
• Malware attacks ○ Ransomware ○ Trojan ○ Worm ○ Spyware ○ Bloatware ○ Virus ○ Keylogger ○ Logic bomb ○ Rootkit	5	Malware Attacks	Evaluating
• Physical attacks ○ Brute force ○ Radio frequency identification (RFID) cloning ○ Environmental	10	Wireless Attacks	Analyzing
• Network attacks ○ Distributed denial-of-service (DDoS) ▪ Amplified ▪ Reflected ○ Domain Name System (DNS) attacks ○ Wireless ○ On-path ○ Credential relay ○ Malicious code	8	Attacks on Networks	Evaluating
• Application attacks ○ Injection ○ Buffer overflow ○ Replay ○ Privilege escalation ○ Forgery ○ Directory traversal	5	Application Vulnerabilities and Attacks	Analyzing
• Cryptographic attacks ○ Downgrade ○ Collision ○ Birthday	3	Cryptographic Limitations and Attacks	Analyzing
• Password attacks ○ Spraying ○ Brute force	7	Types of Authentication Credentials	Applying
• Indicators ○ Account lockout	5	Malware Attacks	Analyzing

◦ Blocked content ◦ Impossible travel ◦ Resource consumption ◦ Resource inaccessibility ◦ Out-of-cycle logging ◦ Published/documented ◦ Missing logs			
2.5 Explain the purpose of mitigation techniques used to secure the enterprise.			
• Segmentation	9	Security Appliances	Applying
• Access control ◦ Access control list (ACL) ◦ Permissions	7	Access Controls	Understanding
• Application allow list	5	Securing Endpoint Devices	Applying
• Isolation	9	Security Appliances	Evaluating
• Patching	5	Hardening Endpoints	Applying
• Encryption	3	Using Cryptography	Analyzing
• Monitoring	9	Security Appliances	Applying
• Least privilege			
• Configuration enforcement			
• Decommissioning			
• Hardening techniques ◦ Encryption ◦ Installation of endpoint protection ◦ Host-based firewall ◦ Host-based intrusion prevention system (HIPS) ◦ Disabling ports/protocols ◦ Default password changes ◦ Removal of unnecessary software	5	Securing Endpoint Devices	Applying
3.0 Security Architecture			
3.1 Compare and contrast security implications of different architecture models.			
• Architecture and infrastructure concepts ◦ Cloud ▪ Responsibility matrix ▪ Hybrid considerations ▪ Third-party vendors ◦ Infrastructure as code ◦ Serverless ◦ Microservices	11	Cloud Computing Security	Applying
◦ Network infrastructure ▪ Physical isolation ▫ Air-gapped ▪ Logical segmentation	9	Secure Infrastructure Design	Understanding
▪ Software-defined networking (SDN) ◦ On-premises ◦ Centralized/decentralized	11	Virtualization Security	Applying
◦ Containerization ◦ Virtualization ◦ IoT ◦ Industrial control systems (ICS)/supervisory control and data acquisition (SCADA) ◦ Real-time operating system (RTOS) ◦ Embedded systems ◦ High availability	6	Embedded Systems and Specialized Devices	Applying
• Considerations ◦ Availability ◦ Resilience ◦ Cost ◦ Responsiveness ◦ Scalability ◦ Ease of deployment			

◦ Ease of recovery ◦ Patch availability ◦ Inability to patch ◦ Power ◦ Compute			
3.2 Given a scenario, apply security principles to secure enterprise infrastructure. • Infrastructure considerations ◦ Device placement ◦ Security zones ◦ Attack surface ◦ Connectivity ◦ Inline ◦ Failure modes ▪ Fail-open ▪ Fail-closed ◦ Device attribute ▪ Active vs. passive ◦ Network appliances ▪ Jump server ▪ Proxy server ▪ Intrusion protection system (IPS)/intrusion detection system (IDS) ▪ Load balancer ▪ Sensors ◦ Port security ◦ Firewall types ▪ Web application firewall (WAF) ▪ Unified threat management (UTM) ▪ Next-generation firewall (NGFW) ▪ Layer 4/Layer 7	9	Security Appliances	Evaluating
• Secure communication ◦ Virtual private network (VPN) ◦ Remote access ◦ Tunneling ▪ Transport Layer Security (TLS) ▪ Internet protocol security (IPSec) ◦ Software-defined wide area network (SD-WAN) ◦ Secure access service edge (SASE) • Selection of effective controls	4 11	Secure Communication and Transport Virtualization Security	Understanding Applying
3.3 Compare and contrast concepts and strategies to protect data. • Data types ◦ Regulated ◦ Trade secret ◦ Intellectual property ◦ Legal information ◦ Financial information ◦ Human- and non-human-readable • Data classifications ◦ Sensitive ◦ Confidential ◦ Public ◦ Restricted ◦ Private ◦ Critical • General data considerations ◦ Data states ▪ Data at rest ▪ Data in transit ▪ Data in use ◦ Data sovereignty ◦ Geolocation	3	Data Controls	Analyzing

• Methods to secure data ○ Geographic restrictions ○ Encryption ○ Hashing ○ Masking ○ Tokenization ○ Obfuscation ○ Segmentation ○ Permission restrictions			
3.4 Explain the importance of resilience and recovery in security architecture. • High availability ○ Load balancing vs. clustering • Site considerations ○ Hot ○ Cold ○ Warm ○ Geographic dispersion • Platform diversity • Multi-cloud systems • Continuity of operations • Capacity planning ○ People ○ Technology ○ Infrastructure • Testing ○ Tabletop exercises ○ Fail over ○ Simulation ○ Parallel processing • Backups ○ Onsite/offsite ○ Frequency ○ Encryption ○ Snapshots ○ Recovery ○ Replication ○ Journaling • Power ○ Generators ○ Uninterruptible power supply (UPS)	13 13	Preparatory Plans Resilience Through Redundancy	Analyzing Applying
4.0 Security Operations			
4.1 Given a scenario, apply common security techniques to computing resources. • Secure baselines ○ Establish ○ Deploy ○ Maintain • Hardening targets ○ Mobile devices ○ Workstations ○ Switches ○ Routers ○ Cloud infrastructure ○ Servers ○ ICS/SCADA ○ Embedded systems ○ RTOS ○ IoT devices • Wireless devices ○ Cryptographic protocols ○ Authentication protocols	8 5 6 9 10	Security Monitoring and Alerting Securing Endpoint Devices Protecting Mobile Devices Security Appliances Wireless Security Solutions	Applying Understanding Understanding Applying Analyzing

▪ Site surveys ▪ Heat maps			
• Mobile solutions ○ Mobile device management (MDM) ○ Deployment models ▪ Bring your own device (BYOD) ▪ Corporate-owned, personally enabled (COPE) ▪ Choose your own device (CYOD) ○ Connections methods ▪ Cellular ▪ Wi-Fi ▪ Bluetooth	6	Protecting Mobile Devices	Understanding
• Application security ○ Input validation ○ Secure cookies ○ Static code analysis ○ Code signing • Sandboxing • Monitoring	6	Application Security	Applying
4.2 Explain the security implications of proper hardware, software, and data asset management. • Acquisition/procurement process • Assignment/accounting ○ Ownership ○ Classification • Monitoring/asset tracking ○ Inventory ○ Enumeration • Disposal/decommissioning ○ Sanitization ○ Destruction ○ Certification ○ Data retention	15	Asset Protection	Applying
4.3 Explain various activities associated with vulnerability management. • Identification methods ○ Vulnerability scan ○ Application security ▪ Static analysis ▪ Dynamic analysis ▪ Package monitoring ○ Threat feed ▪ Open-source intelligence (OSINT) ▪ Proprietary/third-party ▪ Information-sharing organization ▪ Dark web ○ Penetration testing ○ Responsible disclosure program ▪ Bug bounty program ○ System/process audit • Analysis ○ Confirmation ▪ False positive ▪ False negative ○ Prioritize ○ Common Vulnerability Scoring System (CVSS) ○ Common Vulnerability Enumeration (CVE) ○ Vulnerability classification ○ Exposure factor ○ Environmental variables ○ Industry/organizational impact ○ Risk tolerance	12	Vulnerability Scanning	Analyzing

• Vulnerability response and remediation
  ◦ Patching
  ◦ Insurance
  ◦ Segmentation
  ◦ Compensating controls
  ◦ Exceptions and exemptions
• Validation of remediation
  ◦ Rescanning
  ◦ Audit
  ◦ Verification
• Reporting			

4.4 Explain security alerting and monitoring concepts and tools.
• Monitoring computing resources
  ◦ Systems
  ◦ Applications
  ◦ Infrastructure
• Activities
  ◦ Log aggregation
  ◦ Alerting
  ◦ Scanning
  ◦ Reporting
  ◦ Archiving
  ◦ Alert response and remediation/validation
  ▪ Quarantine
  ▪ Alert tuning
• Tools
  ◦ Security Content Automation Protocol (SCAP)
  ◦ Benchmarks
  ◦ Agents/agentless
  ▪ Security information and event management (SIEM)
  ▪ Antivirus
  ▪ Data loss prevention (DLP)
  ◦ Simple Network Management Protocol (SNMP) traps
  ◦ NetFlow
  ◦ Vulnerability scanners
	8	Security Monitoring and Alerting	Analyzing

4.5 Given a scenario, modify enterprise capabilities to enhance security.
• Firewall
  ◦ Rules
  ◦ Access lists
  ◦ Ports/protocols
  ◦ Screened subnets
• IDS/IPS
  ◦ Trends
  ◦ Signatures
• Web filter
  ◦ Agent-based
  ◦ Centralized proxy
  ◦ Universal Resource Locator (URL) scanning
  ◦ Content categorization
  ◦ Block rules
  ◦ Reputation
• Operating system security
  ◦ Group Policy
  ◦ SELinux
• Implementation of secure protocols
  ◦ Protocol selection
  ◦ Port selection
  ◦ Transport method
• DNS filtering
• Email security
	9	Security Appliances	Analyzing

	8	Email Monitoring and	Applying

and Conformance (DMARC) ○ DomainKeys Identified Mail (DKIM) ○ Sender Policy Framework (SPF) ○ Gateway • File integrity monitoring • DLP • Network access control (NAC) • Endpoint detection and response (EDR)/extended detection and response (XDR) • User behavior analytics			
4.6 Given a scenario, implement and maintain identity and access management. • Provisioning/de-provisioning user accounts • Permission assignments and implications • Identity proofing • Federation • Single sign-on (SSO) ○ Lightweight Directory Access Protocol (LDAP) ○ Open authorization (OAuth) ○ Security Assertions Markup Language (SAML) • Interoperability • Attestation • Access controls ○ Mandatory ○ Discretionary ○ Role-based ○ Rule-based ○ Attribute-based ○ Time-of-day restrictions ○ Least privilege • Multifactor authentication ○ Implementations ▪ Biometrics ▪ Hard/soft authentication tokens ▪ Security keys ○ Factors ▪ Something you know ▪ Something you have ▪ Something you are ▪ Somewhere you are • Password concepts ○ Password best practices ▪ Length ▪ Complexity ▪ Reuse ▪ Expiration ▪ Age ○ Password managers ○ Passwordless • Privileged access management tools ○ Just-in-time permissions ○ Password vaulting ○ Temporal accounts	7 7	Types of Authentication Credentials Authentication Best Practices	Analyzing Applying
4.7 Explain the importance of automation and orchestration related to secure operations. • Use cases of automation and scripting ○ User provisioning ○ Resource provisioning ○ Guard rails ○ Security groups ○ Ticket creation ○ Escalation	14	Security Operations	Understanding

◦ Continuous integration and testing ◦ Integrations and Application programming interfaces (APIs) • Benefits ◦ Efficiency/time saving ◦ Enforcing baselines ◦ Standard infrastructure configurations ◦ Scaling in a secure manner ◦ Staff retention ◦ Reaction time ◦ Workforce multiplier • Other considerations ◦ Complexity ◦ Cost ◦ Single point of failure ◦ Technical debt ◦ Ongoing support			
4.8 Explain appropriate incident response activities. • Process ◦ Preparation ◦ Detection ◦ Analysis ◦ Containment ◦ Eradication ◦ Recovery ◦ Lessons learned	13	Incident Investigation	Applying
• Training • Testing ◦ Tabletop exercise ◦ Simulation • Root cause analysis • Threat hunting • Digital forensics ◦ Legal hold ◦ Chain of custody ◦ Acquisition ◦ Reporting ◦ Preservation ◦ E-discovery	13	Preparatory Plans	Applying
4.9 Given a scenario, use data sources to support an investigation. • Log data ◦ Firewall logs ◦ Application logs ◦ Endpoint logs ◦ OS-specific security logs ◦ IPS/IDS logs ◦ Network logs ◦ Metadata • Data sources ◦ Vulnerability scans ◦ Automated reports ◦ Dashboards ◦ Packet captures	13	Incident Investigation	Analyzing
5.0 Security Program Management and Oversight			
5.1 Summarize elements of effective security governance. • Guidelines • Policies ◦ Acceptable use policy (AUP) ◦ Information security policies ◦ Business continuity ◦ Disaster recovery	14	Administration	Applying

◦ Incident response ◦ Software development lifecycle (SDLC) ◦ Change management • Standards ◦ Password ◦ Access control ◦ Physical security ◦ Encryption • Procedures ◦ Change management ◦ Onboarding/offboarding ◦ Playbooks • External considerations ◦ Regulatory ◦ Legal ◦ Industry ◦ Local/regional ◦ National ◦ Global • Monitoring and revision • Types of governance structures ◦ Boards ◦ Committees ◦ Government entities ◦ Centralized/decentralized • Roles and responsibilities for systems and data ◦ Owners ◦ Controllers ◦ Processors ◦ Custodians/stewards			
5.2 Explain elements of the risk management process. • Risk identification • Risk assessment ◦ Ad hoc ◦ Recurring ◦ One-time ◦ Continuous • Risk analysis ◦ Qualitative ◦ Quantitative ◦ Single loss expectancy (SLE) ◦ Annualized loss expectancy (ALE) ◦ Annualized rate of occurrence (ARO) ◦ Probability ◦ Likelihood ◦ Exposure factor ◦ Impact • Risk register ◦ Key risk indicators ◦ Risk owners ◦ Risk threshold • Risk tolerance • Risk appetite ◦ Expansionary ◦ Conservative ◦ Neutral • Risk management strategies ◦ Transfer ◦ Accept ▪ Exemption ▪ Exception ◦ Avoid ◦ Mitigate	15	Risk Management	Analyzing

• Risk reporting
• Business impact analysis
  ◦ Recovery time objective (RTO)
  ◦ Recovery point objective (RPO)
  ◦ Mean time to repair (MTTR)
  ◦ Mean time between failures (MTBF)			

5.3 Explain the processes associated with third-party risk assessment and management.
• Vendor assessment
  ◦ Penetration testing
  ◦ Right-to-audit clause
  ◦ Evidence of internal audits
  ◦ Independent assessments
  ◦ Supply chain analysis
• Vendor selection
  ◦ Due diligence
  ◦ Conflict of interest
• Agreement types
  ◦ Service-level agreement (SLA)
  ◦ Memorandum of agreement (MOA)
  ◦ Memorandum of understanding (MOU)
  ◦ Master service agreement (MSA)
  ◦ Work order (WO)/statement of work (SOW)
  ◦ Non-disclosure agreement (NDA)
  ◦ Business partners agreement (BPA)
• Vendor monitoring
• Questionnaires
• Rules of engagement
	15	Risk Management	Applying
			
			Remembering

5.4 Summarize elements of effective security compliance.
• Compliance reporting
  ◦ Internal
  ◦ External
• Consequences of non-compliance
  ◦ Fines
  ◦ Sanctions
  ◦ Reputational damage
  ◦ Loss of license
  ◦ Contractual impacts
• Compliance monitoring
  ◦ Due diligence/care
  ◦ Attestation and acknowledgement
  ◦ Internal and external
  ◦ Automation
• Privacy
  ◦ Legal implications
  ▪ Local/regional
  ▪ National
  ▪ Global
  ◦ Data subject
  ◦ Controller vs. processor
  ◦ Ownership
  ◦ Data inventory and retention
  ◦ Right to be forgotten
	14	Administration	Understanding

5.5 Explain types and purposes of audits and assessments.
• Attestation
• Internal
  ◦ Compliance
  ◦ Audit committee
  ◦ Self-assessments
• External
  ◦ Regulatory
  ◦ Examinations
	12	Audits and Assessments	Applying

◦ Assessment ◦ Independent third-party audit • Penetration testing ◦ Physical ◦ Offensive ◦ Defensive ◦ Integrated ◦ Known environment ◦ Partially known environment ◦ Unknown environment ◦ Reconnaissance ▪ Passive ▪ Active			
5.6 Given a scenario, implement security awareness practices. • Phishing ◦ Campaigns ◦ Recognizing a phishing attempt ◦ Responding to reported suspicious messages • Anomalous behavior recognition ◦ Risky ◦ Unexpected ◦ Unintentional • User guidance and training ◦ Policy/handbooks ◦ Situational awareness ◦ Insider threat ◦ Password management ◦ Removable media and cables ◦ Social engineering ◦ Operational security ◦ Hybrid/remote work environments • Reporting and monitoring ◦ Initial ◦ Recurring • Development • Execution	15	Risk Management	Understanding